29/04/2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: Jan 2025 - April 2025 Maximum Marks: 100 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01/04 Programme: B. Tech Computer/IT Engineering	Class: SY	Semester: IV (SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: Computer/IT	
Course Code:116U01C401/116U04C401	Name of the Course: Probability, Statistics and Optimization Techniques	
Instructions: 1) All questions are compulsory 2) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four** of the following.	20
i)	Three factories A, B, C produce 30%, 50% & 20% of the total production of an item. Out of their production 80%, 50% & 10% are defective respectively. An item is chosen at random and found to be defective. Using Bayes theorem find the probability that it was produced by the factory A.	5
ii)	The regression lines of a sample are $x + 6y = 6$, and $3x + 2y = 10$. Find (a) Sample means $\bar{x}$ and $\bar{y}$, (b) coefficient of correlation between x and y.	5
iii)	Test the significance of the difference between the means of two normal populations with the same standard deviation from the following data.	5

	Size	Mean	S.D.
Sample I	100	64	6
Sample II	200	67	8

iv)	Convert the given LPP into the standard form Minimise $z = 3x_1 + 2x_2 + 5x_3$ Subject to $2x_1 - 3x_2 \le 3,\ 3x_1 + 2x_2 + 3x_3 \ge 5,\ 3x_1 + 2x_3 \le 2$ Where $x_1, x_2 \ge 0$ and x_3 is unrestricted in sign.	5
v)	A repair shop, attended by a single mechanic, has an average of four customers an hour who bring small appliances for repair The mechanics inspects them for defects and for this he takes six min. on an average. Arrivals are Poisson and service rate has an exponential distribution. You are required to: (a) Find the proportion of time during which there is no customer in the shop (b) Find the probability of finding at least one customer in the shop	5
vi)	The joint probability density of two random variables is given by $f_{XY}(x,y) = \begin{cases} 15e^{-3x-5y}, & x > 0,\ y > 0 \\ 0, & eleswhere \end{cases}$ (a) Find the probability that $1 < X < 2$ *and* $0.2 < Y < 0.3$ (b) Find marginal probability distributions of X and Y.	5

Q2 A	Solve the following.	10
i)	Compute Spearman's rank correlation coefficient between X and Y from the following data. X : 18, 20, 34, 52, 12 Y : 39, 23, 35, 18, 46.	5
ii)	The following table gives the number of accidents in a district during a week. Apply χ^2 – test to find whether the accidents are uniformly distributed over the week	5

Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
No. of accidents	13	12	11	9	15	10	14

OR

Q2 A	Use Lagrange's method to Optimize $z = 2x_1^2 + x_2^2 + 3x_3^2 + 10x_1 + 8x_2 + 6x_3 - 100$ Subject to $x_1 + x_2 + x_3 = 20, \quad x_1, x_2, x_3 \geq 0$	10
Q 2 B	Solve any **One** of the following.	10
i)	A maintenance service facility has Poisson distribution arrival rates, negative exponential service times, and operates on a first come, first served queue discipline. Breakdowns occur on an average of three per day. The maintenance crew can service, on an average, six machines per day. Find the: **(a)** Utilization factor of the service facility **(b)** Mean waiting time in the system **(c)** Mean number of machines in the system	10
ii)	In an intelligence test administered to 1000 students the average was 42 & standard deviation was 24.Find the numbers of students **(a)** exceeding 50 marks **(b)** Between 30&54 marks **(c)** The least score of top 100 students.	10
Q3	Solve any **Two** of the following.	20
i)	**(a)** For a normal variate X with mean 25 and standard deviation 10, Find $P(15 < X < 35)$	05
	(b) The average of marks scored by 32 boys is 72 with standard deviation 8 while that of 36 girls in 70 with standard deviation 6. Test at 1% level of significance whether the Boys perform better than the girls.	05
ii)	A sample of 25 pairs of values of x and y lead to the following results. $\sum x = 127, \sum y = 100, \sum x^2 = 760, \sum y^2 = 449, \sum xy = 500.$ Later on it was found that two pairs of values were taken as (8, 14) and (8, 6) instead of correct values .(8, 12) and (6, 8). Find corrected correlation coefficient between x and y.	10
iii)	Find (a) All basic solutions (b) All feasible basic solutions (c) All degenerate solutions hence decide the optimal feasible basic for the following L.P.P. Maximise $z = 2x_1 + 3x_2 + x_3 + x_4$ Subject to $x_1 - 3x_2 + 2x_3 + x_4 = 5$ $x_1 + x_2 + 3x_3 - 2x_4 = 4$ where $x_1, x_2, x_3, x_4 \geq 0$	10